

Lahari Madhusudhan

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EXPERIENCE

Machine Learning and Autonomy Engineer

Mar 2024- Present

Tocaro Blue, Birmingham, AL

- Owned the end-to-end computer vision pipeline from data collection and labeling coordination to model training, validation, and C++ edge deployment across edge devices
- Implemented efficient probabilistic tracking logic for radar perception systems, improving multi-target continuity across merged detections and noisy observations without adding significant computational overhead
- Designed and implemented map-aware path-planning algorithms using nautical chart data to generate safe navigation routes based on vessel constraints, water depth, fairways, and restricted regions
- Diagnosed sensor alignment and compass bias issues using statistical analysis, improving radar azimuth calibration and overall tracking consistency
- Established structured radar and vision data workflows using Git LFS and DVC, improving dataset versioning, experiment traceability, and reproducibility across the team
- Developed classification and semantic segmentation workflows to evaluate real-time vision-based object detection for future multi-sensor fusion within a radar-driven autonomy stack
- Developed baseline comparison and metric validation pipelines for perception algorithms, automating report generation and creating Confluence documentation to communicate model performance, data quality issues, and labeling guidelines across teams

Software Engineer, Robotics

May 2023-Aug 2023

Honeywell, Pittsburgh, PA.

- Identified a system integration gap preventing AMRs (OTTO 100, 1500, Lifter) from interfacing with the main assembly-line control system; developed and implemented integration modules in Java and Spring Boot to enable the AMR fleet to operate under the same centralized control software, streamlining automation and improving coordination across the line
- Selected to demonstrate the integrated AMR fleet management system to Honeywell's Chairman and senior leadership, showcasing system performance and contributing to strategic visibility for the R&D initiative
- Worked on Honeywell's proprietary ROS2-like software stack in C++ to automate robot performance testing; developed scripts and test modules that benchmarked AMR behavior against existing performance standards, reducing manual testing time and improving validation consistency for the R&D team

Systems Engineer

Dec 2021-Jul 2022

Tata Consultancy Services, Bengaluru, KA, India

- Identified data inconsistencies in legacy telemetry systems and developed Python modules integrating Orion, Piper, and BERT APIs to automate network data acquisition, timestamp alignment, and latency diagnostics, improving real-time analysis accuracy by 40%
- Addressed the time-consuming and error-prone process of locating faulty and legacy network areas by developing statistical and ML-based anomaly detection pipelines that analyzed latency and packet loss, automating fault identification and enabling planners to efficiently optimize new network layouts

EDUCATION

Master of Engineering, Robotics, and Intelligent Autonomous Systems

Aug 2022-Aug 2023

University of Cincinnati, Cincinnati, OH

GPA: 3.8

- Relevant Coursework: Machine Learning, Artificial Intelligence, Industrial AI, Robot control and Design, Decision Engineering, Deep Learning, Industrial AI

Bachelor of Engineering, Mechanical Engineering

Aug 2017-Jun 2021

SCSVMV, Kancheepuram, India

GPA: 9.31

- Relevant Coursework: Engineering Mechanics, Object-Oriented Programming, Kinematics of Machines, Dynamics of Machines, Probability and Statistics, Electronics and Microprocessor Lab

ACADEMIC PROJECTS

Prediction of Material Removal in CMP Process for Semiconductor Fabrication

Jan 2023- Apr 2022

University of Cincinnati, Cincinnati, OH

- Designed a machine learning framework to predict Material Removal Rate (MRR) in the Chemical-Mechanical Planarization (CMP) process using experimental semiconductor fabrication data
- Faced with inconsistent wafer planarity, developed and tuned KNN, SVR, and Random Forest regression models in Python to estimate MRR; Random Forest improved prediction accuracy by 25%, enabling data-driven control and higher process yield.

Fuzzy Bayesian Network "Risk Analysis" on UAV Missions/ Decision Engineering

Aug 2022-Dec 2022

University of Cincinnati, Cincinnati, OH

- Developed a quantitative UAV risk assessment framework integrating Fault Tree Analysis (FTA) and Bayesian Network modeling to identify mission-critical failure modes.
- Addressing the lack of probabilistic safety tools, built a fuzzy Bayesian model in MATLAB to analyze failure dependencies; identified flight control and power failures as top risk contributors, enhancing UAV mission reliability and decision-making.

Autonomous Delivery Robot/ Robotics Control And Design

Aug 2022-Dec 2022

University of Cincinnati, Cincinnati, OH

- The project aimed to enable a Delivery robot to autonomously map and navigate an unknown environment using Simultaneous Localization and Mapping (SLAM) with onboard sensors.
- Configured and executed SLAM (slam_toolbox, gmapping, Gazebo) in ROS Melodic on a Raspberry Pi 4B, fusing LiDAR and IMU data for mapping and path planning; tuned navigation parameters to achieve stable autonomous motion and precise obstacle avoidance in real time.

SKILLS

Software: C++, MATLAB, Python, ROS2, Java, Docker, Git, Linux, Test-Driven Development, Agile Methodology.

Machine Learning: Keras, PyTorch, Ultralytics, Semantic Segmentation, Instance Segmentation, Classification.

Robotics: Computer Vision, Sensor fusion, Data Association, Filtering, Motion Planning.